Solar Al appendix

Sun · Al · downloadable appendix

Document	Solar Al appendix
Subject	Solar photospheric abundance appendix · Al
Author	Ryan Schmitt
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Solar aluminium: 6.469 ± 0.034

Asplund, Amarsi & Grevesse 2021: 6.43 ± 0.03 · this measurement +0.039 dex

Measured the way Asplund, Amarsi & Grevesse (2021) measure it: the Asplund et al. 2021 line set (Nordlander & Lind 2017: the Al I lines of their Table A.1), 6 lines measured on one spectrum, each part weighted by its uncertainty.

Al I: 6.428 ± 0.033 · Kitt Peak (1984, molecfit) · VIS · Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)

Al I: 6.450 ± 0.063 · Kitt Peak (1984, molecfit) · red-optical · Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)

Al I: 6.552 ± 0.043 · Kitt Peak (1984, molecfit) · NIR · Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi)

Highlighted Products

VIS

6.387 ± 0.030

Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · HARPS · Reference Grade

red-optical

6.406 ± 0.033

Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · IAG FTS · Reference Grade

NIR

6.552 ± 0.043

Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · Kitt Peak (1984, molecfit) · Reference Grade

Line grades

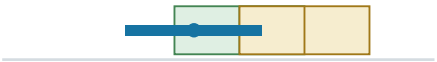
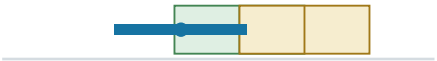
Every product is measured on one of three grades of spectral lines and the grades are never mixed in one number.

Grade	What it means	This element
Reference Grade	lines from a published solar line set (Asplund et al. 2021 and similar), measured with the oscillator strengths the authors used	14 products -- Al I · Asplund et al. 2021 lines
Codex Grade	lines chosen by the Codex whose oscillator strength carries a published uncertainty (laboratory measurement, NIST accuracy class A-C, or the paper's own stated error), between 5% and 60% deep	12 products -- Al I · Codex-graded lines

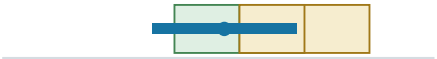
Grade	What it means	This element
Deep Grade	the same oscillator-strength rule on lines deeper than 60% -- stronger lines, more sensitive to damping and the model atmosphere	none for the Sun: every line in this grade is excluded as saturated or an outlier (see Excluded lines)

Error-bar Forest Plot

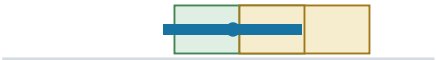
VIS — HARPS
A(Al) dex; axis 6.32–6.52

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.409 ±0.032
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		6.387 ±0.030
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.403 ±0.031
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.381 ±0.030

VIS — Kitt Peak (Kurucz 2005)
A(Al) dex; axis 6.32–6.52

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.451 ±0.030
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		6.429 ±0.031
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.446 ±0.032
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.423 ±0.033

VIS — Kitt Peak (1984, molecfit)
A(Al) dex; axis 6.32–6.52

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines		6.450 ±0.031
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines		6.428 ±0.033
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines		6.450 ±0.031
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines		6.427 ±0.032

VIS axis: 6.32 · 6.37 · 6.42 · 6.47 · 6.52 A(Al) dex

red-optical — IAG FTS
A(Al) dex; axis 6.34–6.56

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines	6.423 ±0.032
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines	6.406 ±0.033
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines	6.423 ±0.032
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines	6.406 ±0.033

red-optical — Kitt Peak (1984, molecfit)
A(Al) dex; axis 6.34–6.56

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines	6.467 ±0.058
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines	6.450 ±0.063
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines	6.467 ±0.058
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines	6.450 ±0.063

red-optical axis: 6.34 · 6.40 · 6.45 · 6.50 · 6.56 A(Al) dex

NIR — CRIRES+ Y (full arm)
A(Al) dex; axis 6.00–7.48

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 1 line	6.747 ±0.702
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 1 line	6.732 ±0.702

NIR — Kitt Peak (1984, molecfit)
A(Al) dex; axis 6.00–7.48

Al I · Asplund et al. 2021 lines · 1D LTE Reference Grade · 2 lines	6.569 ±0.045
Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) Reference Grade · 2 lines	6.552 ±0.043
Al I · Codex-graded lines · 1D LTE Codex Grade · 2 lines	6.569 ±0.045
Al I · Codex-graded lines · 1D NLTE (Amarsi) Codex Grade · 2 lines	6.552 ±0.043

NIR axis: 6.00 · 6.37 · 6.74 · 7.11 · 7.48 A(Al) dex

How to read this forest. Each row is one independent product, preserving its band, holding, engine, indicator and line set. Engines and indicator families are never averaged. Each bar is the uncertainty as Asplund et al. 2021 quote it: line-to-line scatter, the 3D-model term and the NLTE term (at least 0.03 dex). Atomic-data and observational terms are not in the bar, as in Asplund; they stay in the full error budget attached to the row. Grades describe the supplied line-pool provenance, not engine quality; evaluated theoretical CNO oscillator strengths are not relabelled as laboratory evidence. Quarantined, failed and unconstrained fits have no abundance mark. Experimental products remain explicitly not adopted. Literature references provide context, never tuning targets.

How this was measured

Every aluminium value on this page comes from fitting synthetic spectra to the observed solar spectrum, line by line or band by band, with the aluminium abundance as the free parameter. The synthesis includes the atomic lines and the molecular opacity in every window, so blended absorption is modelled rather than attributed to the line being measured. The continuum is placed locally around every line by comparing the observation with the synthesis on its own highest pixels, and ground-based spectra are telluric-corrected across their full range.

Each indicator is measured independently on each solar spectrum and reported separately; nothing is averaged across indicators or instruments. Where a 3D or NLTE correction is published for the lines, it is applied as its own product (1D-LTE, 1D-NLTE and 3D-NLTE are separate rows). Every value carries a full uncertainty budget: fit, line strengths, microturbulence, continuum, fit window, model atmosphere, telluric residual, and the spread between spectra, each measured by refitting with that one input changed. The headline at the top is the Asplund-grade product: a published line set, the most complete treatment, then the most lines.

Band	VIS
What was measured	Al I · Codex-graded lines
Solar spectra	HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE
Lines	2
Result A(Al)	6.381–6.427 (1D-NLTE)
Largest uncertainty terms (dex)	continuum placement 0.07, line strengths (gf) 0.06

Band	VIS
What was measured	Al I · Asplund et al. 2021 lines
Solar spectra	HARPS, Kitt Peak (1984, molecfit), Kitt Peak (Kurucz 2005)
Treatments	1D-LTE, 1D-NLTE
Lines	2
Result A(Al)	6.387–6.429 (1D-NLTE)
Largest uncertainty terms (dex)	continuum placement 0.07, line strengths (gf) 0.06

Band	red-optical
What was measured	Al I · Codex-graded lines
Solar spectra	IAG FTS, Kitt Peak (1984, molecfit)
Treatments	1D-LTE, 1D-NLTE
Lines	2

Band	red-optical
What was measured	Al I · Codex-graded lines
Result A(Al)	6.406–6.450 (1D-NLTE)
Largest uncertainty terms (dex)	continuum placement 0.07, line strengths (gf) 0.05

Band	red-optical
What was measured	Al I · Asplund et al. 2021 lines
Solar spectra	IAG FTS, Kitt Peak (1984, molecfit)
Treatments	1D-LTE, 1D-NLTE
Lines	2
Result A(Al)	6.406–6.450 (1D-NLTE)
Largest uncertainty terms (dex)	continuum placement 0.07, line strengths (gf) 0.05

Band	NIR
What was measured	Al I · Codex-graded lines
Solar spectra	Kitt Peak (1984, molecfit)
Treatments	1D-LTE, 1D-NLTE
Lines	2
Result A(Al)	6.552 (1D-NLTE)
Largest uncertainty terms (dex)	fit window 0.11, continuum placement 0.06

Band	NIR
What was measured	Al I · Asplund et al. 2021 lines
Solar spectra	CRIRES+ Y (full arm), Kitt Peak (1984, molecfit)
Treatments	1D-LTE, 1D-NLTE
Lines	1–2
Result A(Al)	6.552–6.732 (1D-NLTE)
Largest uncertainty terms (dex)	fit window 0.11, continuum placement 0.06

Solar aluminium context

Aluminium is traced by Al I lines in the optical and near-infrared. The reference lines are the six Asplund et al. 2021 used (Nordlander & Lind 2017: 6696, 6698, 7835, 8912, 10768 and 10872 Å) with the oscillator strengths and uncertainties those authors adopted. Al I departs from LTE, so each spectrum also carries a non-LTE-corrected product; each set, spectrum and treatment remains a separate product.

Published line sets · gf uncertainty model · Methodology

Download the evidence

Download PDF — Solar Al appendix

Source feed — includes quarantined fit outputs, not adopted abundances

Complete product visibility audit

Shared website/PDF report

Reproducibility manifest

Complete references and product-to-source bindings

Reproducibility

Generated from the pinned science snapshot. Source commit f5c6b3d35f016dd66b74d3a098459bd648f9677f
Feed Al.json v1.68 · updated 2026-10-29T03:00:31Z
Generated 2026-10-28T21:25:21-06:00 (source commit timestamp)
Feed SHA-256 c5cc06e65a2e5771861960db7708634a890ac640950364de3422e3c0c1bcef9e
Generator scripts/generate_cno_publication.py v1.0.0

Pinned scientific source

References / Data & Model Sources

Sources are separated by their role. Registered grid lineage and quarantined diagnostics do not imply an adopted abundance or an applied correction.

See the source notes for the distinction between reference lineage and the treatment actually recorded in each feed entry.

Science and provenance source notes

Solar abundance and indicator analyses. Solar Al comparison literature; not an adopted Codex abundance. Sources: [1], [2].

Solar abundance and indicator analyses. Published line set AL_AGSS21 (the line selection). Sources: [3].

Atomic data and gf authorities. Al I oscillator strengths of this pool, as the product's own uncertainty budget records them (with their lineage). Sources: [3].

Model and grid sources. Feed treatment: 1D-LTE synthesis (Turbospectrum); no NLTE/3D correction claimed. Sources: [4].

Model and grid sources. Feed treatment: 1D NLTE, per-line correction from the Amarsi 2020 GALAH departure grid via PySME (data/nlte_grids/Al_Amarsi2020_PySME.csv). Sources: [5].

Product-to-source index

NIR · CRIRES+ Y (full arm) · Al I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Al-products-3dced287aa93

NIR · CRIRES+ Y (full arm) · Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-6a4b66082042

NIR · Kitt Peak (1984, molecfit) · Al I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Al-products-e1f71442de77

NIR · Kitt Peak (1984, molecfit) · Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-9443d9b76f0f

NIR · Kitt Peak (1984, molecfit) · Al I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Al-products-2c817b8eea2e

NIR · Kitt Peak (1984, molecfit) · Al I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-fcec422376c8

VIS · HARPS · Al I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Al-products-49c83f2985f1

VIS · HARPS · Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-cfa1cc6e0651

VIS · HARPS · Al I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Al-products-8d6674191138

VIS · HARPS · Al I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-f02ef045fac9

VIS · Kitt Peak (Kurucz 2005) · Al I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Al-products-763c8ec6b825

VIS · Kitt Peak (Kurucz 2005) · Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-4ae1b7f5d552

VIS · Kitt Peak (Kurucz 2005) · Al I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Al-products-fbc7225a6f74

VIS · Kitt Peak (Kurucz 2005) · Al I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-e336923b786f

VIS · Kitt Peak (1984, molecfit) · Al I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Al-products-e748030f2e3a

VIS · Kitt Peak (1984, molecfit) · Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-260a78aecdcd

VIS · Kitt Peak (1984, molecfit) · Al I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Al-products-f0ee54abeefa

VIS · Kitt Peak (1984, molecfit) · Al I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-c71470f7634e

red-optical · IAG FTS · Al I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Al-products-e7e99f42d031

red-optical · IAG FTS · Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-189b8acbdd3b

red-optical · IAG FTS · Al I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Al-products-ca9c883560bd

red-optical · IAG FTS · Al I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-ad4018a70e99

red-optical · Kitt Peak (1984, molecfit) · Al I · Asplund et al. 2021 lines · 1D LTE · [1], [2], [3], [4] Al-products-172960c23449

red-optical · Kitt Peak (1984, molecfit) · Al I · Asplund et al. 2021 lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-c85886319365

red-optical · Kitt Peak (1984, molecfit) · Al I · Codex-graded lines · 1D LTE · [1], [2], [3], [4] Al-products-1be4caee7bb3

red-optical · Kitt Peak (1984, molecfit) · Al I · Codex-graded lines · 1D NLTE (Amarsi) · [1], [2], [3], [5] Al-products-38c43a7a6482

Complete applicable bibliography

Solar abundance and indicator analyses

[1] — Asplund, M.; Amarsi, A. M.; Grevesse, N. (2021). The chemical make-up of the Sun: A 2020 vision. A&A 653, A141 DOI: 10.1051/0004-6361/202140445. source record

[2] — Lodders, K.; Bergemann, M.; Palme, H. (2025). Solar System Elemental Abundances from the Solar Photosphere and CI-Chondrites. Space Science Reviews 221 DOI: 10.1007/s11214-025-01146-w. source record

[3] — Nordlander, T.; Lind, K. (2017). Non-LTE aluminium abundances in late-type stars. A&A 607, A75 DOI: 10.1051/0004-6361/201730427. source record

Model and grid sources

[4] — Gerber, J. M.; Magg, E.; Plez, B.; Bergemann, M.; Heiter, U.; Olander, T.; Hoppe, R. (2023). Non-LTE radiative transfer with Turbospectrum. A&A 669, A43 DOI: 10.1051/0004-6361/202243673. source record

[5] — Amarsi, A. M.; Lind, K.; Osorio, Y.; Nordlander, T.; Bergemann, M.; et al. (2020). The GALAH Survey: non-LTE departure coefficients for large spectroscopic surveys. A&A 642, A62 DOI: 10.1051/0004-6361/202038650. source record